

Gate E4 report — run_v1.2_r1 (Paper IV v1.2)

Verdict: **PASS**. Evidence directory: 20_EVIDENCE/E4/.

E4 — Formal reproduction (isolated project, shared pinned dependencies)

Setup

- **Runner:** `auditor_build.py`, written by the auditor. It does not use Lake or the author's runner. It compiles serially, one lean process at a time with `-j 8`. Each module gets the lakefile `[leanOptions] (pp.unicode.fun=true, relaxedAutoImplicit=false, linter.unusedSimpArgs=false)` plus the owning `lean_lib's moreLeanArgs (-s 32768)`, using Lake's `isBuildableModule` rule. Every command line is recorded at the top of each module log.
- **Toolchain:** `lean.exe` v4.28.0, commit 7e01a1bf, sha256 6a66c4a5...
- **Sources:** a fresh extraction of `LEAN_SOURCE_piv-v12-fb459343d234.zip` into `05_BUILD/main/src` (615/615 manifest hashes checked; no `.lake` directory, no `.olean`).
- **Objects:** written only to `05_BUILD/main/obj`, which started empty. No author `.olean/.ilean` was used.
- **Dependencies:** the shared package build directories (`mathlib`, `plausible`, `LeanSearchClient`, `importGraph`, `proofwidgets`, `aesop`, `Qq`, `batteries`), read-only via `LEAN_PATH`. A snapshot of 113 821 files taken before and after shows 0 changed, 0 added, 0 removed, with the same `Mathlib.olean` sha256 (`cache_diff.json`).
- **Attempts:** attempt 1 (15:07 UTC) failed on the auditor's own option-flattening bug (C-04) and is preserved. Attempt 2 ran 15:09–21:33 UTC (6.4 h).

Results — main cut

- **607/607 modules PASS**, exit 0, with no failed, blocked or skipped module. `records.jsonl` has one row per module, including the source hash, dependency `olean` hashes, the command and the `olean` sha256.
- **All 19 FREEZE_SCOPE targets were executed fresh and PASS**. Each target log is identical to the author's segment-2 log apart from the command line (`target_log_analysis.json`).
- **Bit-for-bit reproduction:** all **607 .olean files are byte-identical** to the `olean` hashes recorded by the author across the two segments (`olean_vs_author.json`). Separately, a 37-module recompilation reproduced identical `olean`s (determinism). Together these independently corroborate the author's two-segment reuse (resolves X-21).
- **Axiom records:** 461 under the author's counting rule, of which 314 are genuine `#print axioms` outputs. Every set is a subset of `{propext, Classical.choice, Quot.sound}`.
- **Sorry and errors:** "sorry" occurs nowhere (case-insensitive) in any of the 608 auditor logs, or in the author's logs. No `error` appears in any target log. The negative control (`AuditorNegControl.lean`) shows that Lean 4.28 prints "declaration uses `sorry`" with backticks. The literal pattern in both runners would have missed it, so the explicit rescan and the axiom cones are the operative checks (C-07, finding X-28).
- **Exports:** `ReleaseExportCheck` (224 `#check` lines, import `PaperIV` only) compiled with 0 errors.

Auditor's independent checks (`AuditorChecks.lean`, `AuditorChecks.log`)

- The elaborated types of the headline declarations were printed; their definitions (`RootedDefectAt`, `CliquePartition`, `IsDefectRoot`, `IsCanonicalPiece`, `defSplitGraph`) match `E3_STATIC_RECORD`.
- `#print axioms` on 18 headline declarations: all exactly `[propext, Classical.choice, Quot.sound]`.
- **Own cone traversal** (type + value, transitive) of 50 public declarations: **0 non-standard axioms, 0 sorryAx, 0 constants under the root Erdos81**.
- **Removal provenance** (`AuditorASProbe.lean`):
 - None of the six named removal keys (`AlonShapira.lemma_4_2`, `near_chordal_of_induced_cycles_little0`, `...few_induced_cycles`, `EditRoute.editApproxAt_all`, `EditRoute.fixedL4Localization_unconditional`, `AFKS.strong_regularity`) occurs in the cones of the public Theorem C (`rooted_defect_eventual/_maximum`), `E35.theoremC_tower*`, `fixed_defect_stability_explicit`, `fixed_defect_joint_stability_real` or `sequence_arbitrary_orders_stability`. This confirms `FDCheck.ASCheck` independently.
 - Shared helpers (`AlonShapira.IsChordal.induce`, `EditRoute.* arithmetic lemmas`) do occur, as §7 discloses.

- **However**, the cones of `E32.cp_classification_of_theoremCPrimeC` and `E32.ref15Theorem11_of_theoremCPrimeC` contain `AlonShapira.lemma_4_2`, `near_chordal_of_induced_cycles_littleO`, `EditRoute.editApproxAt_all`, `EditRoute.fixedL4Localization_unconditional` and `AFKS.strong_regularity`. They enter through `E32.theoremC_at := DefectSharpPublication.rooted_defect_eventual`, the historical existential assembly. Appendix F.4 Table 9 cites `E32.cp_classification_of_theoremCPrimeC` for the "unrestricted equality classification" (Cor 6.3, (6.18)). The formal cp-part of that classification therefore depends on the Alon–Shapira route that §7 and [23] describe as historical and not selected. This is finding X-27 (MODERATE, provenance/conformance). There is no correctness impact: that route is fully proved, with standard axioms.

Results — annex (BoundedCliqueGap)

- Project: a fresh extraction of `LEAN_SOURCE_SNAPSHOT_v1.0.zip` (its lakefile, manifest and toolchain equal the bound freeze files), with the 38 annex files overlaid (38/38 manifest hashes OK), in its own objects directory `05_BUILD/annex/obj`. Run serially after the main build.
- **50/50 modules PASS** for the closure of `BoundedCliqueGap.AxiomCheck`, the same number as the internal report. The audit reports "633 declarations audited; axioms $\subseteq$ [propext, Classical.choice, Quot.sound]; no `Erdos81* name`". `chordal_gap_linear_cliqueFree` and four related declarations depend on exactly the three standard axioms.

Limits

- Same machine and shared Mathlib/third-party oleans. The dependencies were not rebuilt, so this is not a clean-room Mathlib build. The `leanchecker` kernel replay was not performed (C-02 probe aborted).
- The build establishes that the formal statements are proved with standard axioms. It does not establish exposition adequacy (A.2) or manuscript correspondence beyond E3.

E4 verdict: PASS (with the limits above).

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